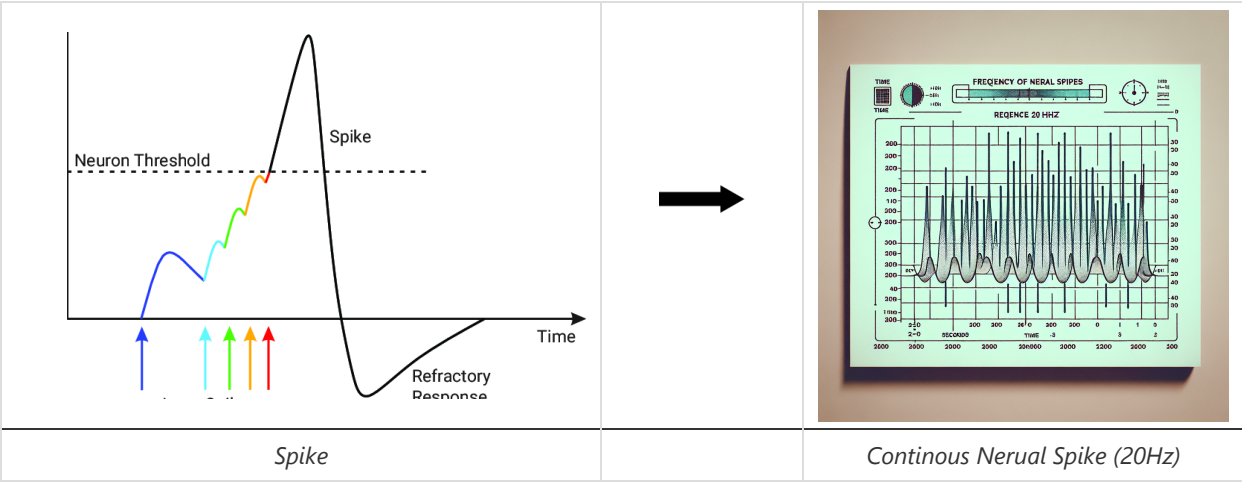


Neural Encoding

In this content, we show how to use information theory to validate simple stimulus–response models of neural coding of dynamic stimuli.

Background

Because these models require specification of spike timing precision, they can reveal which time scales contain information in neural coding. This approach shows that dynamic stimuli can be encoded efficiently by single neurons and that each spike contributes to information transmission.



The *frequency of neural spikes*, often referred to as the firing rate, is a key aspect of neuronal activity. It represents how often a neuron generates an action potential or "spike" over a period of time. This rate can vary widely depending on the type of neuron, the organism, and the specific conditions or stimuli the neuron is responding to.

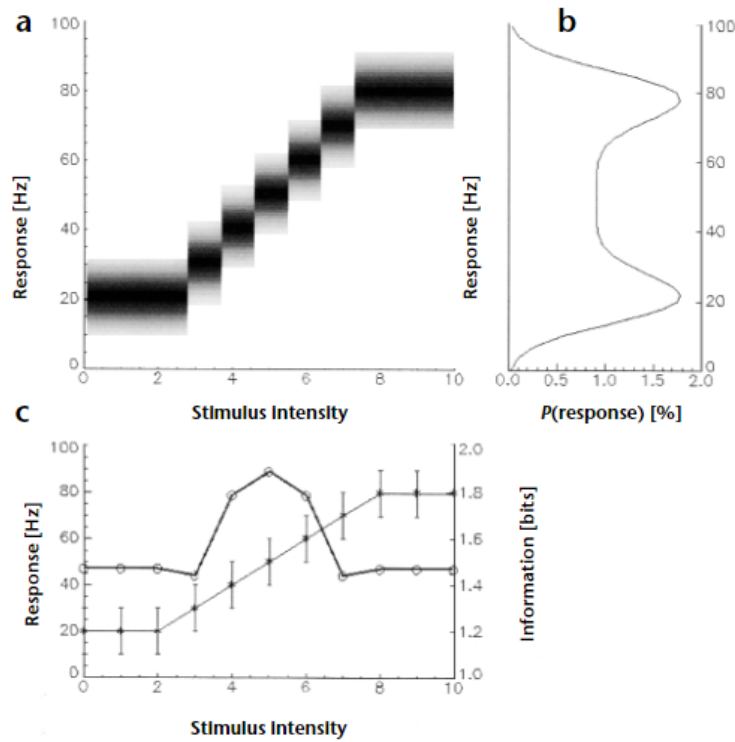
General Concepts of Information Theory

Information theory and significance of neuronal encoding. Information theory measures the statistical significance of how neural responses vary with different stimuli. That is, it determines how much information about stimulus parameter values is contained in neural responses.

The less for the vaildability of the neuron is, the information obtained per trial is greater. If response variability is described by the variance, then neuronal information can be described by the signal detection measure d' , which equals the differential response normalized by response variances. (Gaussian distribution)

Formula	Explanation
$p(r_i)$	Probability that neural response takes the value r_i
$p(s_j)$	Probability that stimulus condition takes the value s_j
$p(r_i s_j)$	Probability that neural response takes the value r_i when stimulus condition s_j is presented (conditional probability)
$I(R, s_x) = \sum_i p(r_i s_j) \log_2 \frac{p(r_i s_j)}{p(r_i)}$	Information about stimulus condition s_x (measure how the distribution of responses to any particular stimulus condition X is different from all other conditional distributions that can be obtained)
$I(R, S) = \sum_i \sum_j p(s_j)p(r_i s_j) \log_2 \frac{p(r_i s_j)}{p(r_i)}$	Average information obtained from all stimulus conditions

- Expirement on a mock neuron



(a) Complete response distributions for each stimulus intensity; darker values indicate higher probabilities. (b) Summing these values along the horizontal lines leads to the overall response probability distribution (right), assuming that each stimulus is equally likely to occur. (c) Information theory allows one to replace the traditional stimulus-response curve (mean $\pm$ s.d.) with an information curve (thick line, $I(R, s_x)$) that indicates how well different values of the stimulus are encoded in the response.

Information theory and neural information transfer. Information theory can be used to calculate maximal rates of information transfer. This measure, which is estimated from the set of all possible neuronal responses, is used to evaluate neuronal precision.

- **The entropy of a distribution of stimulus conditions**

Entropy, which measures the information required to code a variable with a certain probability distribution by characterizing how many states it can assume and the probability of each.

Formula	Explanation
$p(s, r) = p(s r)p(r)$	Bayes' theorem
$H(S) = -\sum_i p(s_i) \log_2 p(s_i)$	Entropy of S
$H(R, S) = -\sum_i \sum_j p(s_i, r_j) \log_2 p(s_i, r_j)$	Joint entropy of R and S
$H(R S) = -\sum_j p(s_j) \sum_i p(r_i s_j) \log_2 p(r_i s_j)$	Conditional entropy of R given S or neuronal noise
$H(S R) = -\sum_j p(r_j) \sum_i p(s_i r_j) \log_2 p(s_i r_j)$	Conditional entropy of S given R or stimulus equivocation

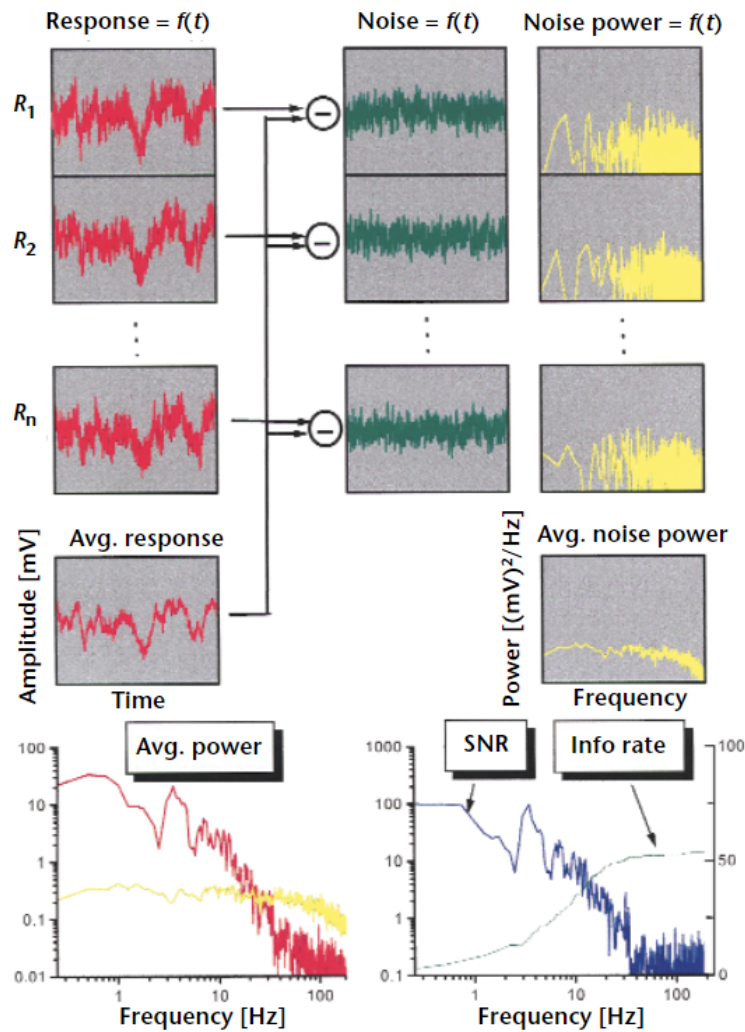
- **Equivalent forms for average information:** (Mutual Information)

$$I(R, S) = H(R) - H(R|S)$$

$$I(R, S) = H(S) - H(S|R)$$

$$I(R, S) = H(R) + H(S) - H(R, S)$$

- **Flow chart of how to measure the channel capacity of a neuron**



The same stimulus is presented n times while the responses R_i are measured (left). These responses are averaged to obtain the average response R_{avg} . The difference between each R_i and R_{avg} become the noise traces N_i (middle). These are Fourier-transformed to the noise power spectra $N_i(f)$ (right), which can be averaged as well. Bottom left, power spectra of the mean response (red) together with the mean power spectra of the noise (yellow). Bottom right, ratio of these two functions, the so-called signal-to-noise ratio or SNR, together with the cumulative information rate. Response and noise data were created in a pseudo-random way from Gaussian distributions

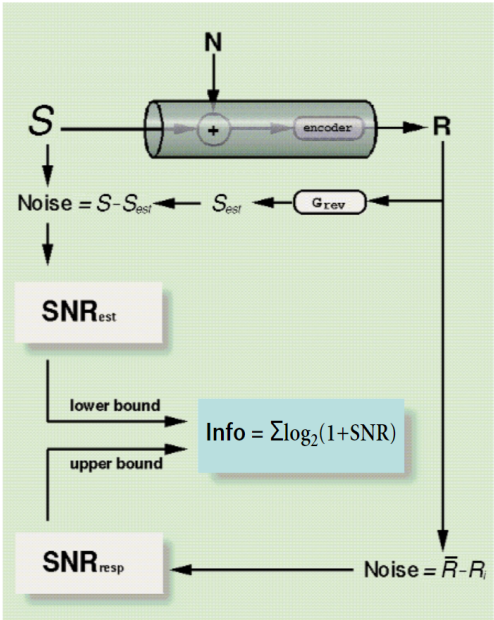
data processing inequality theorem

$$I(S, R_1) \geq I(S, R_2)$$

where S is encoded by a first neuron in a set of neuronal responses R_1 , and R_1 is then encoded by a second set of neuronal responses R_2 .

In practice, it is easier to represent neural responses with minimal assumptions about the neural code than it is to find appropriate stimuli and the correct parameters to describe them. Neural responses can be represented with high temporal precision, and, with enough data, the relationship of any neural response measure to the stimulus conditions can be evaluated. Information measures can then be used to determine the limiting spike-timing precision involved in that particular encoding, for example by calculating the point at which information values stop increasing when analyzed over progressively shorter time windows. Similarly, information theory can guide the choice of parameters to represent the information being tested.

- Summary diagram for calculation of upper and lower bounds on information transfer

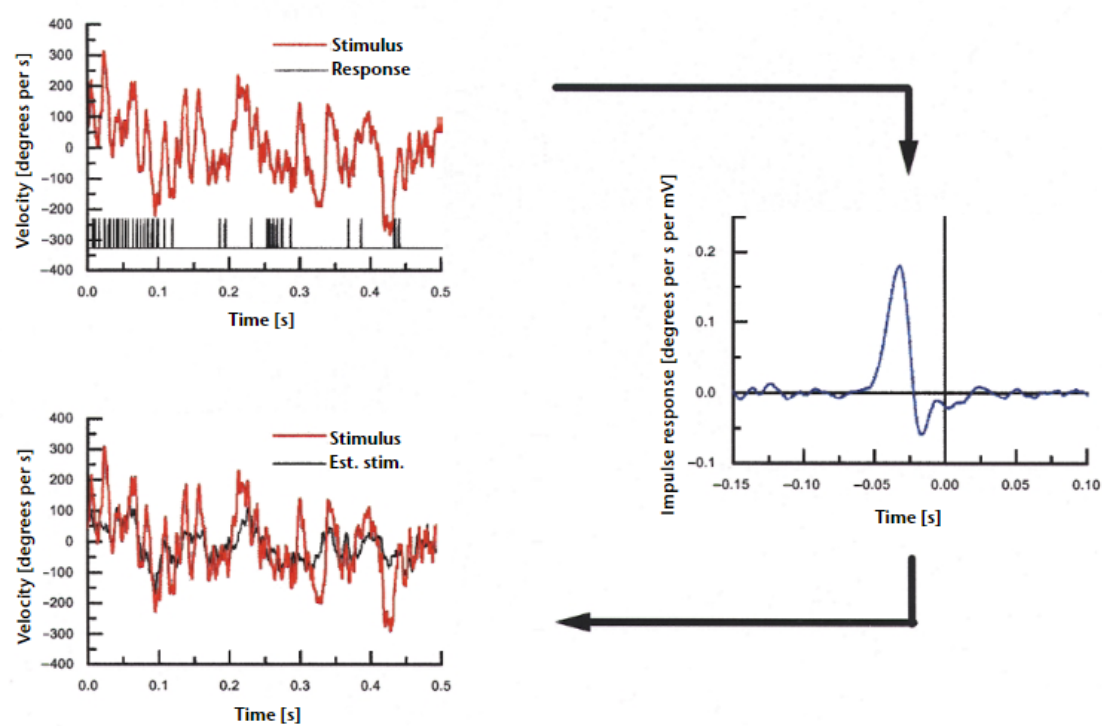


Top, situation where a stimulus S is corrupted by additive noise and subsequently fed through an unknown encoder to result in the response R . The lower bound is obtained with a linear reverse filter operation. The upper bound is obtained directly by comparing average and individual responses.

Dynamic stimuli

Estimating the probabilities of stimulus and response

- Calculates information directly from the neural response by estimating its entropy, $H(R)$, and neural noise, $H(R|S)$.
- With the added assumption that the neuronal response amplitudes, expressed in the frequency domain, have Gaussian probability distributions.
- Calculate information transfer for each possible stimulus condition to obtain the complete curve: assumes a representation (choice of parameters) describing the stimulus conditions and a model relating these stimulus conditions to neural responses.



Linear reconstruction formulas

Formula	Explanation
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Formula	Explanation
$S_{est}(f) = H(f) \cdot R(f)$	S_{est} is obtained by a linear filtering operation on R
$H(f) = \frac{\langle R^*(f) \cdot S(f) \rangle}{\langle R^*(f) \cdot R(f) \rangle}$	$H(f)$
$N = S - S_{est}$	The noise
$SNR = \frac{\langle S_{est} S_{est}^* \rangle}{\langle N N^* \rangle}$	The signal-noise ratio
$\gamma^2 = \frac{\langle S^* R \rangle \langle S^* S \rangle}{\langle R^* S \rangle \langle R^* R \rangle}$	The coherence between S and R
$SNR = \gamma^2 / (1 - \gamma^2)$	The signal to noise ratio
$Info_{LB} = - \int_0^\infty \log_2(1 - \gamma^2) df$	The information

Experiments

Table 1. Methods and assumptions of four ways to calculate neural information.

Method of estimation	Driving principle	Simplifying assumption	Further assumption
Lower bound	Find 'best' S_{est} from R $I(S, R) \rightarrow I(S, S_{est})$	Gaussian S Calculate $N = S - S_{est}$ $\rightarrow$ Equations in Box 3	Linear decoder $\rightarrow$ Box 4 and actual coherence
Absolute lower	Find 'best' S_{est} from R Find smallest $I(S', S_{est})$ that would give the same error as $(S - S_{est})^2$	1) Use Gaussian S $\rightarrow$ Equation 3.1 in ref. 31	
Upper bound (when all assumptions are true)	Separate R into a deterministic and a random component by repeating S many times $I(S, R) \rightarrow I(R, R_{det})$	Additive noise: $R_{det} = R_{avg}$ $N = R - R_{avg}$ $I(R, R_{det}) \rightarrow I(R, R_{avg})$	If N is Gaussian, $\rightarrow$ Box 3 and expected coherence
Direct	Separate R into a deterministic and a random component by repeating S many times $I(S, R) \rightarrow I(R, R_{det})$	None except temporal resolution	

S = stimulus, S_{est} = estimated stimulus, R = response, R_{avg} = average response, R_{det} = deterministic part of the response, N = noise. All four methods reformulate the problem of finding the mutual information between S and R , $I(S, R)$, as an equivalent information calculation that is easier to perform.

Table 2. Neural information and spike precision in response to dynamic stimuli.				
Animal system (Neuron) Stimulus	Method	Bits per second	Bits per spike (efficiency)	High-freq. cutoff or limiting spike timing
Fly visual ¹⁰ (HI) Motion	Lower	64	~1	~2 ms
Fly visual ¹⁵ (HI) Motion	Direct	81	—	0.7 ms
Fly visual ³⁷ (HS, graded potential) Motion	Lower and upper	36 104	—	—
Monkey visual ¹⁶ (area MT) Motion	Lower and direct	5.5 12	0.6 1.5	~100 ms
Frog auditory ³⁸ (Auditory nerve) Noise and call	Lower	Noise 46 Call 133	Noise 1.4 (~20%) Call 7.8 (~90%)	~750 Hz
Salamander visual ⁵⁰ (Ganglion cells) Random spots	Lower	3.2	1.6 (22%)	10 Hz
Cricket cercal ⁴⁰ (Sensory afferent) Mechanical motion	Lower	294	3.2 (~50%)	> 500 Hz
Cricket cercal ⁵¹ (Sensory afferent) Wind noise	Lower	75–220	0.6–3.1	500–1000 Hz
Cricket cercal ^{11,38} (10-2 and 10-3) Wind noise	Lower	8–80	Avg = 1	100–400 Hz
Electric fish ¹² (P-afferent) Amplitude modulation	Absolute lower	0–200	0–1.2 (~50%)	~200 Hz

Paper_Note